Use case	Registration	
Release (id)	UC1, v0.1, 19/04/2026	
Context of Use	User authentication and account management.	
Scope	Let an anonymous user to create a new account.	
Level	Primary task	
Primary actor	Anonymous user	
Stakeholder & interest	**Stakeholder**	**Interest**
	Anonymous user	Wants to create an account
Preconditions	The user must be logged out.	
Minimal Guarantees	No duplicated or unsecure account is created.	
Success Guarantees	The anonymous user successfully created an account and is logged in.	
Trigger	The anonymous user selects “register”.	
Description	**Step**	**Action**
	1	The anonymous user selects the functionality “registration”.
	2	The system asks the user to insert username and email.
	3	The user inserts the required data.
	4	The system checks whether an account with at least one between username and email already exists.
	5	The system asks the user to insert the password twice.
	6	The user inserts the required data.
	7	The system checks whether the password satisfies the security requirements and if the two passwords coincide.
	8	The system asks the user’s consent for personal data processing.
	9	The anonymous user accepts the personal data processing.
	10	The system saves the data in the account archive.

	11	The system notifies the user of successful registration with an email.
	12	The system switches to the authenticated user functionalities.
Extension	**Step**	**Other action**
	4a	Invalid username or email: UC1.1
	7a	Invalid password: UC1.2
Technology & data Variations	None	

Alternative flow	Registration: Invalid username or email	
Release (id)	UC1.1, v0.1, 19/04/2026	
Context of Use	User authentication and account management.	
Scope	The system alerts the anonymous user that she has inserted an invalid username or email.	
Level	Subfunction	
Primary actor	Anonymous user	
Stakeholder & interest	**Stakeholder**	**Interest**
	Anonymous user	Wants to create an account
Preconditions	The user inserted an already existing username or email.	
Minimal Guarantees	Duplicated account is not created.	
Success Guarantees	None	
Trigger	UC1	
Description	**Step**	**Action**
	1	The sequence of events starts at step 2 of the main flow (UC1).
	2	The system alerts the user that has inserted an invalid username or email.
Technology & data Variations	None	

Alternative flow	Registration: Invalid password	
Release (id)	UC1.2, v0.1, 19/04/2026	
Context of Use	User authentication and account management.	
Scope	The system alerts the anonymous user who is registering a new account that she has inserted an invalid password.	
Level	Subfunction	
Primary actor	Anonymous user	
Stakeholder & interest	**Stakeholder**	**Interest**
	Anonymous user	Wants to create an account
Preconditions	The user inserted a password that does not satisfy the security requirements or the two passwords do not coincide.	
Minimal Guarantees	Duplicated account is not created.	
Success Guarantees	None	
Trigger	UC1	
Description (Main Success Scenario)	**Step**	**Action**
	1	The sequence of events starts at step 5 of the main flow (UC1).
	2	The system alerts the user that has inserted an invalid password, specifying the problem.
Technology & data Variations	None	

Use case	Account deletion	
Release (id)	UC2, v0.1, 19/04/2026	
Context of Use	Account management.	
Scope	Let an authenticated user to delete her account.	
Level	Primary task	
Primary actor	Passenger or train company	
Stakeholder & interest	**Stakeholder**	**Interest**
	Passenger	Wants to delete her account
	Train company	Wants to delete its account
Preconditions	The user must be logged in.	
Minimal Guarantees	None	
Success Guarantees	The user successfully deleted her account and is logged out.	
Trigger	The anonymous user selects “delete account”.	
Description	**Step**	**Action**
	1	The anonymous user selects the functionality “delete account”.
	2	The system removes the data in the account archive.
	3	The system notifies the user of successful deletion with an email.
	4	The system switches to the unauthenticated user functionalities.
Extension	**Step**	**Other action**
	-	-
Technology & data Variations	None	

Use case	Account data view	
Release (id)	UC3, v0.1, 19/04/2026	
Context of Use	Account management.	
Scope	Let an authenticated user to view her account's data.	
Level	Primary task	
Primary actor	Authenticated user (passenger, train company or network manager)	
Stakeholder & interest	**Stakeholder**	**Interest**
	Authenticated user	Wants to view her account's data
Preconditions	The user must be logged in.	
Minimal Guarantees	None	
Success Guarantees	The user successfully views her account's data.	
Trigger	The anonymous user selects "view account's data".	
Description	**Step**	**Action**
	1	The anonymous user selects the functionality "view account's data".
	2	The system shows the data saved in the account archive.
Extension	**Step**	**Other action**
	-	-
Technology & data Variations	None	

Use case	Account data change	
Release (id)	UC4, v0.1, 19/04/2026	
Context of Use	Account management.	
Scope	Let an authenticated user to change her account's data.	
Level	Primary task	
Primary actor	Authenticated user (passenger, train company or network manager)	
Stakeholder & interest	**Stakeholder**	**Interest**
	Authenticated user	Wants to change her account's data
Preconditions	The user must be logged in.	
Minimal Guarantees	No duplicated or unsecure account is created with a modification.	
Success Guarantees	The authenticated user successfully changed her account's data.	
Trigger	The authenticated user selects "change data".	
Description	**Step**	**Action**
	1	The authenticated user selects the functionality "change data".
	2	The system asks the user which data to modify.
	3	If the user selects username
	3.1	The system asks the user to insert the new username.
	3.2	The user inserts the new username.
	3.3	The system checks whether an account with the same username already exists.
	4	If the user selects email
	4.1	The system asks the user to insert the new email.
	4.2	The user inserts the new email.
	4.3	The system checks whether an account with the same email already exists.
	5	If the user selects password
	5.1	The system asks the user to insert the password twice.

	5.2	The user inserts the required data.
	5.3	The system checks whether the password satisfies the security requirements and if the two passwords coincide.
	6	The system saves the data in the account archive.
	7	The system notifies the user of successful data change with an email.
Extension	**Step**	**Other action**
	4a	Invalid username or email: UC1.1
	7a	Invalid password: UC1.2
Technology & data Variations	None	

Alternative flow	Account data change: Invalid username	
Release (id)	UC4.1, v0.1, 19/04/2026	
Context of Use	Account management.	
Scope	The system alerts the authenticated user that she has inserted an invalid username.	
Level	Subfunction	
Primary actor	Authenticated user	
Stakeholder & interest	**Stakeholder**	**Interest**
	Authenticated user	Wants to change username
Preconditions	The user inserted an already existing username.	
Minimal Guarantees	Duplicated account is not created.	
Success Guarantees	None	
Trigger	UC4	
Description	**Step**	**Action**
	1	The sequence of events starts at step 3.1 of the main flow (UC4).
	2	The system alerts the user that has inserted an invalid username.
Technology & data Variations	None	

Alternative flow	Account data change: Invalid email	
Release (id)	UC4.2, v0.1, 19/04/2026	
Context of Use	Account management.	
Scope	The system alerts the authenticated user that she has inserted an invalid email.	
Level	Subfunction	
Primary actor	Authenticated user	
Stakeholder & interest	**Stakeholder**	**Interest**
	Authenticated user	Wants to change email
Preconditions	The user inserted an already existing email.	
Minimal Guarantees	Duplicated account is not created.	
Success Guarantees	None	
Trigger	UC1	
Description	**Step**	**Action**
	1	The sequence of events starts at step 4.1 of the main flow (UC4).
	2	The system alerts the user that has inserted an invalid username or email.
Technology & data Variations	None	

Alternative flow	Registration: Invalid password	
Release (id)	UC4.3, v0.1, 19/04/2026	
Context of Use	Account management.	
Scope	The system alerts the authenticated user that she has inserted an invalid password.	
Level	Subfunction	
Primary actor	Authenticated user	
Stakeholder & interest	**Stakeholder**	**Interest**
	Authenticated user	Wants to change password
Preconditions	The user inserted a password that does not satisfy the security requirements or the two passwords do not coincide.	
Minimal Guarantees	Duplicated account is not created.	
Success Guarantees	None	
Trigger	UC1	
Description (Main Success Scenario)	**Step**	**Action**
	1	The sequence of events starts at step 5.1 of the main flow (UC4).
	2	The system alerts the user that has inserted an invalid password, specifying the problem.
Technology & data Variations	None	

Use case	Login	
Release (id)	UC5, v0.1, 20/04/2026	
Context of Use	User authentication.	
Scope	Let an anonymous user to login.	
Level	Primary task	
Primary actor	Anonymous user	
Stakeholder & interest	**Stakeholder**	**Interest**
	Anonymous user	Wants to login
Preconditions	The user must be logged out.	
Minimal Guarantees	No user login in an account she has no authorization for.	
Success Guarantees	The anonymous user is logged in her account.	
Trigger	The anonymous user selects “login”.	
Description	**Step**	**Action**
	1	The anonymous user selects the functionality “login”.
	2	The system asks the user to insert username and password.
	3	The user inserts the required data.
	4	The system checks whether both the username and password correspond to an existing account.
	5	The system switches to the authenticated user functionalities.
Extension	**Step**	**Other action**
	4a	Invalid username or password: UC5.1
Technology & data Variations	None	

Alternative flow	Login: Invalid username or password	
Release (id)	UC5.1, v0.1, 20/04/2026	
Context of Use	User authentication.	
Scope	The system alerts the anonymous user that she has inserted an invalid username or password.	
Level	Subfunction	
Primary actor	Anonymous user	
Stakeholder & interest	**Stakeholder**	**Interest**
	Anonymous user	Wants to login
Preconditions	The user inserted a pair of username and password that does not correspond to any existing account	
Minimal Guarantees	No user login in an account she has no authorization for.	
Success Guarantees	None	
Trigger	UC1	
Description (Main Success Scenario)	**Step**	**Action**
	1	The sequence of events starts at step 2 of the main flow (UC5).
	2	The system alerts the user that has inserted an invalid username or password, specifying the problem.
Technology & data Variations	None	

Use case	Logout	
Release (id)	UC6, v0.1, 20/04/2026	
Context of Use	User authentication.	
Scope	Let an authenticated user to logout.	
Level	Primary task	
Primary actor	Authenticated user	
Stakeholder & interest	**Stakeholder**	**Interest**
	Authenticated user	Wants to logout
Preconditions	The user must be logged in.	
Minimal Guarantees	None	
Success Guarantees	The authenticated user is logged out.	
Trigger	The anonymous user selects “logout”.	
Description	**Step**	**Action**
	1	The anonymous user selects the functionality “logout”.
	5	The system switches to the anonymous user functionalities.
Extension	**Step**	**Other action**
	-	None
Technology & data Variations	None	

Use case	Ticket display	
Release (id)	UC7, v0.1, 21/04/2026	
Context of Use	Ticketing management.	
Scope	Let a user view the details of a ticket.	
Level	Primary task	
Primary actor	Authenticated user (passenger, train company or network manager) or anonymous user	
Stakeholder & interest	**Stakeholder**	**Interest**
	Authenticated user	Wants to view tickets' details
	Anonymous user	Wants to view tickets' details
Preconditions	The user must be in an interface where is displayed a ticket. ???fix?	
Minimal Guarantees	None	
Success Guarantees	The authenticated user successfully view a ticket's details.	
Trigger	The authenticated user selects "view tickets' details".	
Description	**Step**	**Action**
	1	The authenticated user selects the functionality "view tickets' details".
	2	The system displays the ticket's details.
Extension	**Step**	**Other action**
	-	None
Technology & data Variations	None	

Use case	Ticket purchase	
Release (id)	UC8, v0.1, 21/04/2026	
Context of Use	Ticketing management.	
Scope	Enable a passenger to purchase a ticket.	
Level	Primary task	
Primary actor	Passenger	
Stakeholder & interest	**Stakeholder**	**Interest**
	Passenger	Wants to buy a ticket
	Payment gateway	Manages transaction
Preconditions	The passenger must be logged in. The passenger must be in an interface displaying the ticket's details.	
Minimal Guarantees	The passenger who could not pay did not purchase any ticket.	
Success Guarantees	The passenger successfully purchased a ticket.	
Trigger	The passenger selects "Buy ticket".	
Description	**Step**	**Action**
	1	The passenger selects the functionality "Buy ticket".
	2	The system asks the user confirmation for the purchase.
	3	The system sends a transaction request to the payment gateway.
	4	The payment gateway manages the transaction.
	5	The payment gateway sends the transaction's status to the system.
	6	If the transaction is not successful
	6.1	The system alerts the passenger about the failed transaction.
	7	Otherwise
	7.1	The system alerts the passenger about the completed transaction.

	7.2	The system notifies the passenger of successful purchase with an email.
	7.3	The system saves the ticket in the account archive.
Extension	**Step**	**Other action**
	-	None
Technology & data Variations	None	

Use case	Ticket cancellation	
Release (id)	UC9, v0.1, 21/04/2026	
Context of Use	Ticketing management.	
Scope	Enable a passenger to cancel purchased a ticket.	
Level	Primary task	
Primary actor	Passenger	
Stakeholder & interest	**Stakeholder**	**Interest**
	Passenger	Wants to buy a ticket
	Payment gateway	Manages transaction
Preconditions	The passenger must be logged in. The passenger must be in an interface displaying the ticket's details. The passenger must have already purchased the ticket.	
Minimal Guarantees	None	
Success Guarantees	The passenger successfully cancelled a purchased aticket.	
Trigger	The passenger selects "Cancel ticket".	
Description	**Step**	**Action**
	1	The passenger selects the functionality "Cancel ticket".
	2	The system asks the user confirmation for the cancellation.
	3	The system sends a transaction request to the payment gateway.
	4	The payment gateway manages the transaction.
	5	The payment gateway sends the transaction's status to the system.
	6	If the transaction is not successful
	6.1	The system alerts the passenger about the failed transaction.
	7	Otherwise

	7.1	The system alerts the passenger about the completed transaction.
	7.2	The system notifies the passenger of successful purchase with an email.
	7.3	The system saves the ticket's cancellation information in the account archive.
Extension	**Step**	**Other action**
	-	None
Technology & data Variations	None	

Use case	Ticket history	
Release (id)	UC10, v0.1, 21/04/2026	
Context of Use	Ticketing management.	
Scope	Enable a passenger to view her tickets' history, both purchased and cancelled.	
Level	Primary task	
Primary actor	Passenger	
Stakeholder & interest	**Stakeholder**	**Interest**
	Passenger	Wants to view tickets' history
Preconditions	The passenger must be logged in.	
Minimal Guarantees	None	
Success Guarantees	The passenger successfully view tickets' history.	
Trigger	The passenger selects "Tickets' history".	
Description	**Step**	**Action**
	1	The passenger selects the functionality "Ticket's history".
	2	The system shows the list of tickets purchased and cancelled by that passenger.
Extension	**Step**	**Other action**
	-	None
Technology & data Variations	None	